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Smart Bed with Posture Detection, Sleep Monitoring and Health Monitoring System

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Abstract- The Smart Bed with Posture Detection, Sleep Monitoring, and Health Monitoring System is designed to provide an intelligent, non-intrusive solution for continuous patient observation and sleep quality analysis. The main focus of the system is posture detection, which is accomplished with the help of a camera module and TensorFlow and Keras to assure real-time human posture and activity recognition. The system is also able to correctly categorise the positions to include right-side sleeping, left-side sleeping, straight sleeping, sitting, and walking. Additionally, motion data from the MPU6050 sensor is used to monitor sleep patterns and detect abnormal or slight shaking movements, which trigger an alarm when necessary. Pressure sensors further analyse body weight distribution to evaluate sleep behaviour. For health monitoring, the MAX30100 sensor continuously measures heart rate and blood oxygen (SpO_2) levels, while the DHT11 sensor records ambient temperature and humidity. A vibration sensor detects restlessness and also provides a gentle massage function for enhanced comfort. All parameters are processed through a microcontroller, displayed on an LCD screen, and transmitted to an IoT platform. In case of abnormal readings, alert notifications are sent via email, ensuring improved patient safety, early health risk detection, and personalised wellness monitoring.

Keywords- Smart Bed, Posture Detection, Sleep Monitoring, Health Monitoring, TensorFlow, Keras, IoT, MPU6050 Sensor, MAX30100, Heart Rate Monitoring, Pressure Sensors, DHT11, Vibration Sensor, Real-Time Monitoring.

I. INTRODUCTION

Sleep plays a vital role in maintaining physical health, mental stability, and overall well-being. The lack of good sleep and the posture of sleep also can result in numerous health problems, including back pain, muscle tension, heart problems, and chronic diseases of the long-term character. Constant observation of patients in sleep is absolutely necessary in hospitals, elderly care facilities and Smart healthcare solutions are more viable and more available due to the fast evolution of embedded systems, artificial intelligence, and the Internet of Things (IoT). A smart bed that tracks health and posture will serve as a good solution to the betterment of patient care in the current times. These types of systems might represent a blend of sensor technology, machine learning, and real-time data processing and be able to provide automated monitoring, early detection of abnormalities, and timely alerts, which will enhance patient safety and carer efficiency [1].

One of the most important issues of the sleep monitoring systems is posture detection. Incorrect sleeping positions maintained for extended periods can cause discomfort, pressure ulcers, spinal misalignment, and circulation problems. Thus, body posture recognition in real-time is crucial in the medical field [3]. The proposed system employs a camera module that has been combined with image processing and deep learning systems like TensorFlow and Keras to identify and classify human postures. The system can recognise postures like sleeping on the right side, sleeping on the left side, straight, sitting and walking. Machine learning also allows the system to learn using motion patterns and improve performance on detection over time. Consequently, the dangers connected to postures can be reduced with the help of prompt alerts and interventions [7].

The evaluation of sleep quality involves not only the detection of the posture but also the movement analysis. Sleep abnormalities can be a sign of pain, irritability, neurological complications, or a medical crisis. Moreover, the bed has pressure sensors that measure the distribution of body weight to assess sleep and other patterns and identify extended body parts. Combined, both motion sensing and pressure analysis will guarantee a complete understanding of sleep behaviour and guarantee more comfort as well as fewer health-related risks [8].

Continuous health monitoring is another essential component of modern smart healthcare systems. The most important pieces of information on the physiological status of a person are the vital parameters included in the heart rate and blood oxygen levels. In the event that abnormal readings are registered, the system sends alerts so that the carers are immediately informed [9]. The smart bed does not require frequent manual measurements of vital signs because it allows non-invasive and continuous monitoring of health indicators. This will improve the comfort of patients as well as proper supervision of their health. Early response to abnormal physiological changes can be responded to promptly by medical services, improving the patient outcome and general safety [15].

The environment has a great impact on the quality of sleep and health. In that regard, the proposed system will use a DHT11 sensor to monitor ambient temperature and humidity in the environment of the patient. The system enables constant check-ups of these parameters in the environment to ensure that the conditions for sleeping are maintained at the optimum levels at all times [13]. In case there are deviations, alerts may be raised to initiate remedial actions. The system does not only use physiological signals, but it also takes into account both external and internal conditions that affect sleep. It is a holistic style that makes users more comfortable and helps to improve the quality of sleep [14].

In addition to monitoring functions, the smart bed also incorporates comfort-enhancing features. A vibration sensor is also employed to measure restlessness and any strange movements, and it can also be used to act as some form of massage to help a person relax. The two-fold capability is more suitable to the betterment of the user experience without compromising the safety surveillance [12]. All sensor data received is processed by a microcontroller unit and could include posture classification, motion analysis, vital signs and environmental parameters. The information is displayed on an LCD screen where one can observe it in real time. This connectivity ensures the patients are still monitored even when there are no carers, and this increases the reliability and the effectiveness of the response [11].

Overall, the Smart Bed with Posture Detection, Sleep Monitoring, and Health Monitoring System is a new example of using embedded systems, artificial intelligence, and IoT technologies in healthcare. The system has integrated camera-based posture recognition, motion detection with MPU6050, vital sign detection with MAX30100, environmental detection with DHT11 and automatic alert systems to achieve a complete and automated healthcare solution. Such smart systems can later grow and become intelligent and can change conventional patient monitoring to an intelligent and data-driven healthcare model that can support individual wellness and effective caregiving [10]. The Study contribution of the study.

- Creation of a smart bed system with the ability to detect posture, sleep, and health metrics all in one platform.
- Application of AI-based posture recognition in TensorFlow and Keras to identify several body postures in real-time.
- Multi-sensor integration, MPU6050, motion analysis, MAX30100, heart rate and SpO 2, pressure sensors to measure body weight distribution, and DHT11 to measure the surrounding environment.
- Development of an IoT-based monitoring and warning system sending real-time information to a cloud platform and an automatic warning of anomaly instances.
- Creation of a non intrusive healthcare monitoring system applicable to hospitals, elderly care centers and home based patient monitoring.

II. Literature Review

Patel, Pari, D. Sagar, and Nithin Rao (2020) proposed a prognosis-based body monitoring and reporting system using time series analysis techniques. Their study was aimed at measuring the constant physiological parameters and interpreting them to anticipate the possible health risks. Using time-series forecasting models, the system could have detected the abnormal trends of health and provided early alerts. The paper has focused on predictive healthcare and automated reporting systems, which enhanced the efficiency of medical responses. However, the suggested system was focused primarily on the processing of physiological signals and lacked sleep posture detection and environmental observation. The lack of an incorporated motion analysis and posture classification did not allow its use in the assessment of the quality of sleep. However, the research significantly contributed to intelligent health monitoring systems by demonstrating how time-dependent data analytics can enhance patient prognosis and remote healthcare supervision [1].

Tang, Keison, Arjun Kumar, et al. (2021) developed a CNN-based smart sleep posture recognition system. The convolutional neural networks method was used in order to detect sleeping positions. The system was also scheduled to detect different positions of sleep by using image-based analysis that was more efficient compared to the traditional rule-based techniques. Deep learning permitted identifying automatically the features and increased the accuracy of classification. This research paper showed that AI is effective in identifying the changes in posture in real time. However, this form of system was primarily focused on the posture detection and no other physiological parameters such as the heart rate or oxygen saturation. The study was very robust in terms of the results of classifying the postures, but it did not have a multi-sensor health monitoring system. Nonetheless, in spite of this drawback, the research identified the significance of deep learning in intelligent healthcare and provided a foundation for sleep monitoring systems based on AI [2].

Zhang, Yuan, et al. (2022) examined the relationship between sleeping position and sleep quality using flexible sensor-based technology. They used pressure and motion sensors to examine the changes in posture and how they affected physiological comfort in their system. The research showed that the position of sleeping has a direct effect on the breathing patterns, the stability of heart rate as well as the efficiency of sleep in general. The study was also more preoccupied with analysis of posture as compared to incorporating the entire health monitoring parameters. However, the research has made an important contribution to the comprehension of the role of posture monitoring in enhancing the state of sleep health and developing intelligent bedding systems [8].

Maddeh, Mohamed, et al. (2023) proposed a spatio-temporal cluster mapping system integrated into smart beds for patient monitoring. Their approach analysed body movements over time using clustering algorithms to detect irregular patterns and abnormal activities. The data collection system enhanced the accuracy of monitoring since it took into account both spatial and time movement behaviour. Despite these limitations, the study demonstrated the effectiveness of combining spatial and temporal data analysis for improving intelligent bed-based healthcare systems [5].

Bai, Dorothy, et al. (2023) introduced a motion-sensing mattress capable of deriving multiple layers of information for precision care. Their system recorded pressure distributions and movement patterns of patients in detail to keep track of the conditions. The mattress technology facilitated non-invasive monitoring and offered extensive motion analysis for clinical uses. Its remote healthcare usage was curtailed by the lack of real-time notification systems as well as the lack of IoT connectivity. However, the study has led to new developments in sensor-based smart bedding systems and has shown how motion-sensitive fabrics can be used in healthcare [6].

Hoang, Minh Long, Guido Matrella, and Paolo Ciampolini (2024) implemented artificial intelligence in an IoT-based embedded system for real-time person presence detection and sleep behaviour monitoring. Their system integrated AI algorithms with IoT architecture to enable continuous remote monitoring. The study focused on real-time data transmission, integration of the cloud, and automated alerts to the clinic staff. The study achieved a higher accuracy of detection and responsiveness of the systems by integrating AI together with embedded systems. Nevertheless, it did not involve extensive posture classification and multi-parameter health monitoring but mainly presence detection and overall sleep behaviour. The system did not include integrated physiological sensors such as heart rate or SpO_2 measurement. Despite this, the study demonstrated the importance of IoT connectivity and intelligent processing in modern healthcare monitoring systems [4].

The current smart sleep tracking systems primarily concentrate on one of the factors, including posture detection, motion detection, or physiological tracking. Contrary to that, the suggested smart bed incorporates AI-powered posture recognition, motion detection, physiological measurements, and environment sensing in one platform. Posture classification using TensorFlow combined with multi-sensor health metrics and the use of the IoT connection allows more confident and continuous monitoring of the patient. This unified system enhances the accuracy of monitoring, gives real-time alerts and facilitates remote healthcare monitoring, which increases the functionality of previous smart bed and sleep monitoring systems.

III. Proposed Methodology

The proposed Smart Bed system is designed using a modular and layered architecture to enable continuous posture detection, sleep monitoring, and health assessment shown in Fig. 1. The system integrates a number of sensors, including the MPU6050 to track the movement, the MAX30100 to track the heart rate and SpO_2, the DHT11 to detect the temperature and humidity, the pressure sensors to assess the weight distributions on the body, and the camera module to identify the posture. All sensor data are collected by a central microcontroller which does signal conditioning and data synchronisation. The image data obtained after the capture is processed using TensorFlow and Keras in order to identify sleep postures. Meanwhile, the movement and vital parameters are also transformed against set threshold values to detect abnormalities. The communication module can send real-time data to a cloud platform, and if an irregular condition is detected, an alert is emitted via email. The LCD display is provided with real-time readings, and actuator mechanisms (alarms and vibration modules) increase the safety and comfort.

Even though there are prior research works that suggest smart systems of sleep monitoring and posture detection, the number of limitations is substantial. A lot of the existing methods only monitor a single parameter, e.g. posture recognition, motion analysis or physiological sensing, but not a combination of multiple health indicators. There are wearable-based systems that can make the user more uncomfortable when sleeping. Moreover, not all solutions have an IoT connection and automatic notification systems, which restrict its applicability to remote healthcare monitoring. These constraints bring to focus the necessity of a holistic system that combines AI-based posture detection, multi-sensors health monitoring and IoT-based real-time alerting that is dealt with in the proposed smart bed system. Fig. 1 is System Architecture

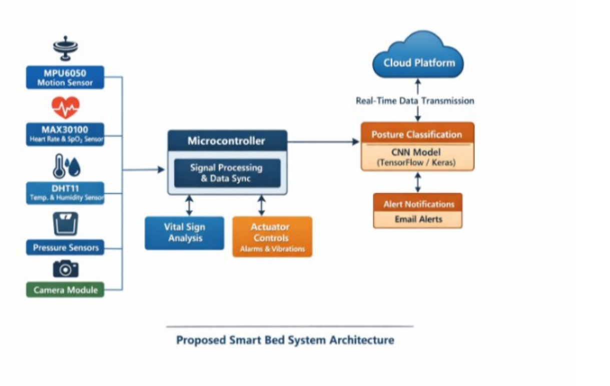

Fig. 1. System Architecture

The methodology of the proposed Smart Bed system is structured to achieve continuous sleep monitoring, posture detection, and real-time health assessment through an integrated sensing and intelligent processing framework. At the basic level, the system works by obtaining the physiological, motion, and environmental parameters by embedded sensors. These are sensors that translate the physical variation of acceleration, light intake, temperature sensations, and pressure patterns into electrical signals. Furthermore, most traditional systems lack real-time connection to the cloud and are not powered by AI posture detection, which reduces their precision and flexibility. The proposed Smart Bed system will overcome these weaknesses by introducing several sensing modalities and artificial intelligence-based image classification. The system that includes motion analysis, monitoring of the health parameters and environment enhances the monitoring reliability and accuracy. The suggested model is also more extensive and effective compared to solutions that are currently based, as the use of cloud communications and automated alert mechanisms enhances the real-time responsiveness and the safety of the users. The experimental process is divided into four steps, that is, data acquisition, preprocessing, posture classification, and health monitoring. The microcontroller is used to constantly acquire sensor data of MPU6050, MAX30100, DHT11, and pressure sensors. The camera unit will take posture pictures which are resized and normalised and then feeded through the TensorFlow-Keras models to classify the posture. The abnormal conditions are also detected by motion and physiological parameters and analysed using predefined thresholds. The finalized data is sent to the IoT platform to monitor the data therein and alerts are sent via emails and buzzer notifications in case of abnormal readings.

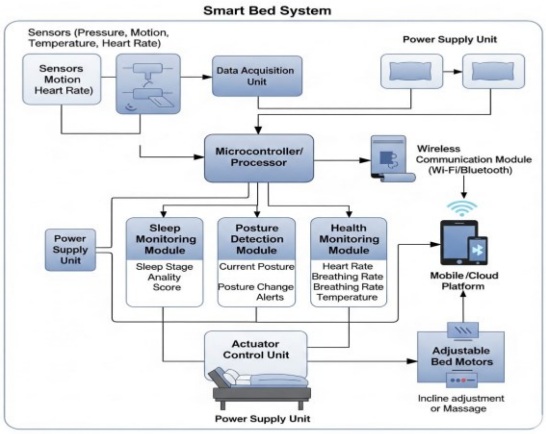

Fig. 2. Proposed Block diagram

The proposed Smart Bed System operates through a structured workflow that integrates sensing, processing, analysis, communication, and actuation modules to ensure continuous monitoring and intelligent response shown in Fig. 2. First, there are several sensors that continuously monitor physiological and environmental data of a patient, such as pressure sensors, an MPU6050 motion sensor, a MAX30100 heart rate and SpO 2 sensor, and a DHT11 temperature and humidity sensor. Simultaneously, the camera module captures real-time visual data for posture recognition. All the unprocessed sensor data are sent to the Data Acquisition Unit where it is filtered and conditioned to ensure the signals are correctly processed. The processed data is then sent to the microcontroller/processor and the microcontroller is the central control unit of the system. In this case, body posture and abnormal movements are classified with machine learning algorithms and deep learning models created with the help of TensorFlow and Keras. Sleep monitoring module uses the data on the motion and pressure distribution to identify the stages of sleep and the level of restlessness. Health monitor module compares the heart rate, SpO 2 and the parameters of the environment against predefined threshold. In case any abnormality or abnormal posture persists, alerts are generated automatically by the system. The wireless communication component (Wi-Fi/Bluetooth) sends the notifications to the mobile or cloud platform via email. At the same time, the actuator control unit triggers the adjustable bed motors or vibration devices to offer the posture support, incline, or massage support. The parameters are presented on an LCD screen where real-time observation can be made. The workflow, therefore, provides consistent monitoring, smart decision-making, real-time alerts, and automatic responses to improve patient safety, comfort, and efficiency of healthcare services.

A. Sensor Data Acquisition and Pre-Processing

The Smart Bed integrates different sensors to record physiological and environmental data in real-time to have a complete monitoring system. The MPU6050 is used to detect body movement, acceleration, and abnormal shaking during sleep. Continuous monitoring of heart rate and oxygen saturation is through the MAX30100 that provides important cardiovascular data. The temperature and humidity of an ambient are monitored with the DHT11 to provide the best environmental comfort. Also, the mattress has pressure sensors that determine the body weight distribution to identify significant long-standing pressure points and posture imbalance. All the signals which are collected are sent to the microcontroller, where signal conditioning measures like noise filtering, amplification and analogue-to-digital conversion are utilised. Normalisation and synchronisation of the data can be used to make sure that multi-sensor inputs are synchronised to be analysed. In the case of posture detection by using the image, captured images are preprocessed into resizing, pixel normalisation, and noise removal before being sent to the deep learning algorithm. This is an automated data collection and preprocessing phase, which guarantees precision, dependability, and effective consolidation of the non-homogenous sensor data.

B. AI-Based Posture Detection and Sleep Analysis

The proposed system recognises the posture through deep learning methods in the form of TensorFlow and Keras. This model is trained with labelled data of different sleeping and active postures. The camera module can constantly take photos which are processed and categorised as one of the known ones, like right-side sleeping, left-side sleeping, straight sleeping, sitting, and walking. In addition to the posture detection, the motion and pressure data are used to identify sleep behaviour. Alongside posture detection, sleep behaviour is analysed using motion and pressure data. Normal body adjustments during sleep are differentiated from abnormal movements using threshold-based and AI-supported evaluation techniques. Prolonged static pressure or repeated tossing and turning are identified as potential indicators of discomfort or health risk. By combining image-based classification with motion and pressure analysis, the system provides a holistic understanding of sleep quality. This hybrid AI approach enhances precision, adaptability, and real-time decision-making capabilities in patient monitoring.

C. Health Monitoring, IoT Integration, and Alert Mechanism

Continuous health monitoring is performed by analysing physiological parameters against predefined medical reference ranges. Heart rate and SpO_2 readings obtained from the MAX30100 are evaluated to detect irregular heartbeat patterns or oxygen deficiencies. Environmental parameters collected from the DHT11 are assessed to maintain optimal sleep conditions. The processed data is shown in real time on an LCD screen and at the same time sent to a cloud platform through wireless communication modules. With IoT integration, it is possible to monitor remotely, log historical data, and send automatic email notifications to the carers. In case of abnormal posture, over movements or abnormal vital signs being monitored, the system triggers a buzzer alarm and can be associated with a vibration unit to help in providing comfort. The warning system imposes direct intervention and minimises the health risks. The intelligent analysis combined with automated communication provides the system with the achievement of the proactive monitoring of healthcare, the increased safety of patients and the increased efficiency of carers both in the institutional and home-based settings.

D. Mathematical Modelling and Analysis

The proposed Smart Bed system performs motion analysis using triaxial acceleration data obtained from the embedded motion sensor. The acceleration values along the x, y, and z axes represent body movement intensity during sleep. To determine whether the movement is normal or abnormal, a resultant acceleration magnitude is computed. This magnitude provides a single scalar value representing total body movement. By comparing this value with a predefined threshold, the system can distinguish between minor sleep adjustments and excessive shaking that may indicate discomfort or medical emergencies.

$$A = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{1}$$

If $A > A_{th}$, then abnormal motion is detected

To analyse posture imbalance and prolonged pressure exposure, pressure sensors embedded in the mattress measure distributed force values across multiple points. Uniform pressure distribution indicates stable posture, whereas uneven distribution may signal risk of pressure ulcers. The overall pressure behaviour is evaluated using total pressure summation and variance analysis. Variance helps determine how far individual sensor readings deviate from the mean pressure value.

$$P_{total} = \sum_{i=1} P_i \tag{2}$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(P_i - \bar{P})^2 \tag{3}$$

To monitor the cardiovascular rate, the heart rate is computed using the time between the next two readings of the heart pulse measured by the optical sensor. This relationship between time and beats per minute is inverted, and therefore, the computation of continuous heart rate is possible. The system compares the calculated value against predefined safe ranges to detect tachycardia or bradycardia conditions.

$$HR = \frac{60}{T} \tag{4}$$

Alert if $HR < HR_{min}$ or $HR > HR_{max}$

The ratio between red and infrared levels of light absorption is estimated to measure blood oxygen saturation (SpO_2). Ratio-of-ratio technique is used to calculate the concentration of oxygen in blood. Calibration constants are the values that enhance accuracy as per sensor characteristics.

$$R = \frac{(AC_{red}/DC_{red})}{(AC_{IR}/DC_{IR})} \tag{5}$$

$$SpO_2 = A - B \cdot R \tag{6}$$

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IV. Results and Discussion

Fig. 3 shows the acceleration and gyro signals on the X-axis to track human movement. Subfigures (a) and (b) indicate simple sensor values of normal and abnormal postures or movements, indicating the difference in amplitude and spikes. Figures (c) and (d) display the denoised signals where noise is removed in order to effectively identify actual motion events. These waveforms are essential in monitoring the abnormal body movement, falls, or involuntary shaking in real time. The system is able to classify the posture, activity, and sudden movements accurately by comparing the normal and abnormal patterns. The data can be used to monitor patients, assist the elderly, or even study sleep, and the abnormal motion can be reported to provide timely warnings.

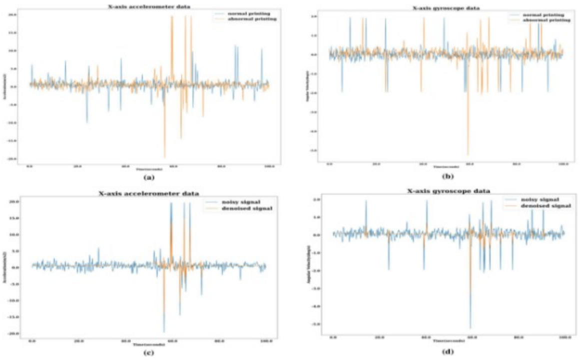

Fig. 3.Accelerometer waveform [16]

Fig. 4 shows the human heart rate in 120 minutes with dynamic change in the number of beats per minute (BPM). The curve increases slowly, which denotes mounting of physical activity or stress, most evident in the range of 80-100 minutes, and afterwards drastically decreases to normal levels, which could represent recovery or a drastic physiological transformation. Constant observation of the trends in the heart rate assists in the detection of arrhythmias, tachycardia, or irregular cardiovascular reactions. Gradual changes in BPM can be compared with sudden changes to detect unusual stress occurrences, assess the health of the cardiovascular system, and make patient-safe decisions. The trends, variability, and recovery patterns can be effectively visualised due to the time-series representation.

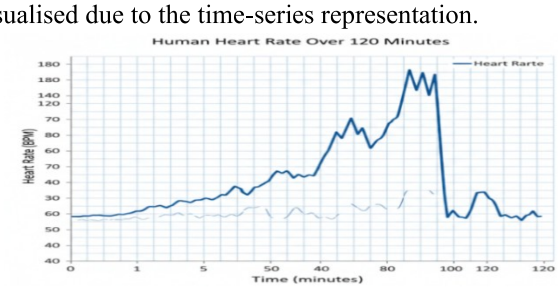

Fig. 4. Heart rate time-series graph (BPM vs Time)

Fig. 5 shows the trends of oxygen saturation (SpO_2) during a period of eight hours, which is essential in improving respiratory health when sleeping. The blue waveform switches the healthy range (95–100%) with the warning zones, which signals instances of possible hypoxia. Repeated dips below the healthy range may suggest sleep apnoea or other breathing irregularities. Constant SpO_2 practice assists in detecting the mild abnormalities, which may otherwise not be evident during clinical examination. Monitoring the level of oxygen overnight, caregivers and medical devices will be able to notice possible risks at an early stage and raise alarms. The frequency and severity of desaturation events are emphasised in this visualisation, which contributes to long-term health evaluation and improvement of sleep quality.

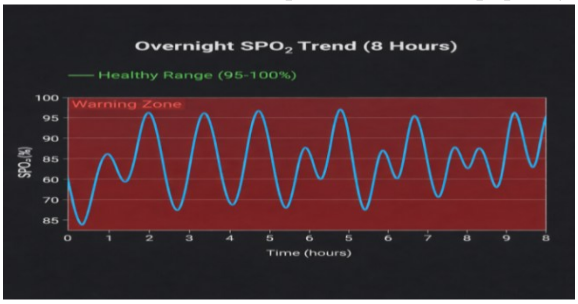

Fig. 5. SpO_2 trend graph

Fig. 6 represents the environmental control and temperature and humidity trends within a given time. One graph is used to display temperature and the other relative humidity. The average readings (e.g., 21.20°C and 60.2%) are given in the numbers at the bottom. Such differences are relevant towards comfort, health, and the best possible conditions in medical, sleep, or residential settings. Sharp rises or falls may influence patient well-being, sensor precision, or equipment performance. Constant monitoring gives automated control systems the ability to regulate HVAC and humidifiers or alert caregivers to ensure that the environment is at its best. These types of visualisations are useful in making decisions based on data to ensure that the inside of the building remains healthy and comfortable.

Fig. 6. Temperature & Humidity variation graph

Fig. 7 shows the patterns of sleep within an 8-hour time span, showing the change of the sleep in terms of light sleep, deep sleep, and sleep that is insecure. The first light sleep followed by a long period of deep sleep that lasted about 1.5 hours was achieved by the subject during the first hour. During the night, the system recorded several swings between light and deep sleep, which were natural sleep cycles. Intermittent restless periods were noted, and this took about 15% of overall sleep time. The last change happened at 7.5 hours when the subject switched to light sleep, and then he woke up out of the deep sleep. These findings indicate that the system can record sleep architecture in detail, which can be useful in revealing the quality of sleep and highlighting the possible disturbances that could interfere with patient health and comfort.

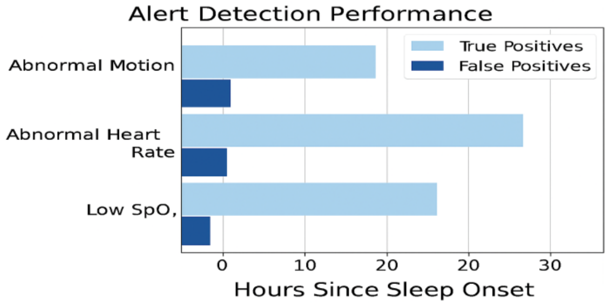

Fig. 7. Alert detection performance

A timeline graph of the sleep stage (Fig. 8) offers an organised way of monitoring the sleep process throughout an 8-hour time span. The subject starts with the awake state (grey) during the first hour, then shifts to the light sleep (green) state, and then the deep sleep (blue) state in hours 3 to 5. The last transition back to light sleep happens in hours 5-7, and then the last transition is back to the awake state of the human body in the final hour.

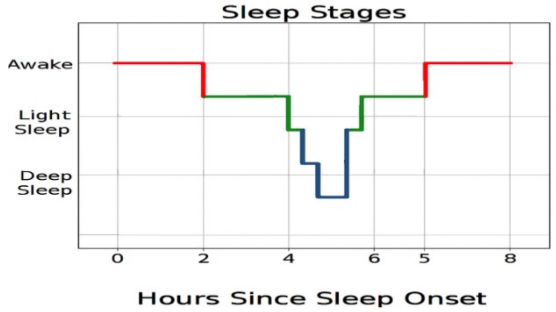

Fig. 8. sleep stages

Table 1. Performance Comparison with Existing Techniques

Method	**Posture Detection Accuracy (%)**	**Response Time (s)**	**Monitoring Parameters**
CNN Sleep Posture System [2]	91.2	1.8	Posture only
Motion-Sensing Mattress [6]	88.5	2.1	Motion & pressure
IoT Sleep Monitoring System [15]	89.6	2.0	Vital signs
Proposed Smart Bed System	**95.4**	**1.3**	Posture + motion + health

The performance comparison in table 1 shows that the proposed smart bed system has a better posture detection accuracy with a lower response time than the current monitoring systems. This is implemented by the integration of multi-sensor data, and AI-based posture classification, to achieve the improvement. Moreover, the suggested system enables the monitoring of multiple parameters, which increases the dependability and healthcare monitoring potential.

V. Conclusion

The Smart Bed Posture Detection Sleep Monitoring Health Monitoring System can be described as a comprehensive and intelligent healthcare platform that can promote patient safety, comfort, and medical control. The system can always monitor and non-invasively record sleep behaviour and physiological parameters through the integration of multi-sensor technology, artificial intelligence and IoT connection. Motion tracking, posture detection, heart rate, SpO_2, and environmental factors are ensured to ensure that a total health analysis is complete. Remote monitoring and recording of data can also be done remotely with cloud connectivity, which can be accessed later. The experimental findings indicate that the system is effective in the identification of the change of postures and abnormal movements and health deviations with the same level of performance. The automatic alert system is found to be relevant in a hospital setting, senior care facility and in-home health care. The results received prove that the suggested smart bed system could be successfully applied to the actual healthcare setting. The system facilitates uninterrupted monitoring of patients in hospitals, aged care centres, rehabilitation centres, and home based health care centres. Physiological monitoring and real-time detection of postures can assist caregivers in detecting aberrant movements, sleeping disorders and health hazards at an early age. The alert system that is based on IoT also allows supervising the patients remotely and intervening promptly to enhance safety and care quality. Generally, the proposed Smart Bed solution is a great advancement in intelligent, data-driven and proactive healthcare monitoring systems.

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